

MARSHALL AAF, KS, USA

WMO: 724550

Lat: 39.050N

Lon: 96.767W

Elev: 1065

StdP: 14.14

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13947

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.5	8.7	-5.8	4.2	7.5	-1.2	5.4	11.3	24.9	42.1	22.4	39.1	6.0	350	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	100.0	75.1	96.3	75.3	93.4	74.8	79.0	91.4	77.7	90.3	76.7	89.0	12.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.9	141.0	84.5	74.4	133.9	83.6	73.1	128.0	82.7	43.3	91.2	42.0	90.3	40.7	88.9	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.7	21.0	18.8	-4.2	103.9	5.0	2.8	-7.8	105.9	-10.7	107.5	-13.5	109.1	-17.1	111.1		
			-4.8	81.4	4.7	1.6	-8.2	82.6	-11.0	83.6	-13.6	84.5	-17.1	85.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.2	30.1	33.8	46.6	55.4	65.5	76.7	80.3	78.7	70.5	57.2	45.1	32.7
	DBStd	19.99	10.84	12.17	11.55	9.55	8.47	6.40	6.20	6.32	8.53	9.51	9.81	10.57
	HDD50	2128	617	469	202	53	3	0	0	0	0	38	204	542
	HDD65	4866	1081	873	579	311	102	2	0	1	40	276	599	1002
	CDD50	4383	1	17	96	215	485	800	941	889	616	262	56	5
	CDD65	1645	0	0	7	23	119	352	476	425	207	35	1	0
	CDH74	18671	0	8	112	306	1217	3964	5708	4797	2153	387	19	0
	CDH80	8791	0	0	18	76	407	1837	3015	2424	911	102	1	0
Wind	WSAvg	8.2	7.4	8.3	9.4	10.1	8.5	8.8	7.3	7.1	7.8	7.8	8.3	7.4
Precipitation	PrecAvg	32.4	0.7	1.1	2.2	3.0	4.6	5.0	4.7	3.9	2.8	2.2	1.4	0.9
	PrecMax	50.1	1.5	2.8	4.6	8.4	13.3	9.2	14.0	6.5	5.0	5.5	3.7	3.0
	PrecMin	23.2	0.0	0.0	0.0	0.9	1.1	0.9	1.6	0.6	0.3	0.2	0.0	0.0
	PrecStd	6.0	0.4	0.7	1.2	1.6	3.2	2.0	3.1	1.6	1.3	1.3	1.0	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.7	74.1	82.2	86.9	92.4	100.1	104.9	103.2	97.5	88.4	76.7	65.4
		MCWB	51.6	53.5	60.5	65.3	70.2	74.2	74.3	75.4	73.9	68.7	59.5	55.2
	2%	DB	58.0	66.4	77.1	81.7	88.1	95.9	101.0	99.0	93.1	82.6	71.3	59.7
		MCWB	46.7	51.4	58.6	63.0	69.1	75.3	75.0	75.8	74.0	64.9	57.7	49.1
	5%	DB	52.4	60.2	72.0	77.2	84.6	93.0	96.9	95.4	89.6	78.1	66.1	54.8
		MCWB	43.1	50.1	55.8	61.2	67.9	74.9	75.6	75.3	73.1	63.2	54.7	46.1
	10%	DB	47.2	54.5	66.9	72.9	81.4	89.9	93.3	91.8	85.7	74.2	61.5	49.7
		MCWB	40.3	45.4	54.0	58.9	66.9	74.1	75.4	74.5	71.3	61.5	51.8	42.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.3	58.9	65.7	68.3	74.4	79.6	80.5	81.1	78.1	72.4	63.1	58.7
		MCDB	58.3	64.8	77.0	79.0	84.8	92.7	92.7	94.7	89.8	81.7	72.8	62.8
	2%	WB	47.5	54.8	62.1	66.0	72.2	77.9	79.0	78.9	76.3	69.6	59.6	51.7
		MCDB	56.0	62.6	70.8	77.1	82.8	90.0	91.8	90.8	88.3	77.2	66.1	55.6
	5%	WB	43.9	50.1	58.7	63.8	70.5	76.5	77.7	77.5	74.6	66.7	56.4	46.5
		MCDB	51.4	62.0	69.5	73.9	80.7	88.6	90.7	89.9	86.2	74.5	64.9	53.4
	10%	WB	40.7	44.5	54.9	61.1	68.9	75.1	76.6	76.2	72.9	63.4	52.8	43.0
		MCDB	47.4	53.5	64.7	70.5	78.4	87.1	89.2	88.2	83.4	70.9	61.0	49.8
Mean Daily Temperature Range	5% DB	MDBR	21.9	22.5	24.3	23.5	21.4	21.0	21.3	21.4	22.2	24.3	22.9	20.4
		MCDBR	30.7	32.9	33.0	29.5	25.6	23.7	25.6	25.8	24.1	28.8	28.8	28.3
		MCWBR	20.3	20.5	17.4	15.6	11.1	8.9	7.9	9.1	9.9	14.7	16.6	19.1
	5% WB	MCDBR	28.6	31.4	26.8	25.1	21.7	21.3	21.6	21.7	21.4	24.1	24.3	25.3
		MCWBR	19.5	20.8	15.5	15.2	10.7	9.7	8.8	9.2	9.8	14.5	16.3	19.0
Clear-Sky Solar Irradiance	taub	0.287	0.302	0.333	0.379	0.411	0.422	0.428	0.427	0.383	0.342	0.302	0.282	
	taud	2.487	2.453	2.410	2.306	2.287	2.290	2.298	2.298	2.374	2.465	2.509	2.502	
	Ebn at Noon	285	294	294	285	276	272	269	267	273	275	276	278	
	Edh at Noon	24	29	34	40	42	42	41	40	35	29	24	23	
All-Sky Solar Radiation	RadAvg	741	984	1291	1637	1870	2120	2088	1863	1539	1125	813	636	
	RadStd	64	103	96	101	148	124	141	95	98	118	68	61	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page